A GRAAL schedule for an unknown weak-Minty parameter

Abstract

We give a predetermined parameter schedule for the ordinary two-line golden ratio algorithm when a global Lipschitz bound is known but the weak-Minty parameter is not. The method uses one operator evaluation per update and retains one averaging-memory orbit. For every admissible weak-Minty parameter strictly below the reciprocal Lipschitz bound, both state sequences remain bounded and the residual squares are summable with the schedule weights. A logarithmic schedule gives a best squared-residual bound of order $\log(N)/N$, with an explicit, deliberately coarse parameter-dependent constant. The proof controls the actual state before invoking boundedness and then bounds all changes in the energy coefficients. No favorable uniform dependence near the parameter boundary or point-convergence conclusion is asserted.

Keywords: golden ratio algorithm; weak Minty inequality; parameter schedule; residual convergence

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1 Model and implemented schedule

Let $F : \mathbb{R}^d \to \mathbb{R}^d$ be globally L-Lipschitz for a known $L > 0$. Assume there exist z^* and ρ with

$$F(z^*) = 0, \qquad \langle F(z), z - z^* \rangle \geq -\frac{\rho}{2}\|F(z)\|^2 \quad (z \in \mathbb{R}^d), \qquad 0 \leq r := \rho L < 1.$$

Neither ρ nor z^* is provided to the algorithm. Put $G = F/L$. Choose any positive, nonincreasing deterministic sequence $(e_k)_{k\geq 0}$ satisfying

$$e_k \leq \tfrac{1}{2}, \qquad e_k \longrightarrow 0, \qquad \sum_{k=0}^{\infty} e_k = \infty.$$

Set $\phi_k = 1 + e_k$, $\eta_k = (1 - e_k)^2$ and $\alpha_k = \eta_k/L$. Starting at an arbitrary z_0 with $\bar{z}_{-1} = z_0$, perform

$$\bar{z}_k = \frac{e_k}{1 + e_k} z_k + \frac{1}{1 + e_k} \bar{z}_{k-1}, \qquad z_{k+1} = \bar{z}_k - \eta_k G(z_k), \qquad k \geq 0. \tag{1}$$

These are the usual two GRAAL lines with the displayed varying parameters. There is one evaluation of F per update, with no restart, parallel run, parameter test or estimate of ρ.

For fixed $\phi > 1$, Alacaoglu, Böhm and Malitsky prove a weak-Minty residual bound with $\alpha = (2 - \phi)/L$ subject to $(2 - \phi)/(\phi L) > \rho$, and ask about choosing the parameter without this information [1, Theorem 7 and subsequent discussion]. The present coupled

steps differ from that displayed fixed-step choice, so the varying schedule needs a direct argument. Adaptive methods are already available in this literature [2]; the claim here concerns exactly (1) on the stated range, with known L.

Theorem 1. *Under these assumptions both (z_k) and $(\bar{z}_k)$ are bounded, and*

$$\sum_{k=0}^{\infty} e_k \|F(z_k)\|^2 < \infty, \qquad \liminf_{k\to\infty} \|F(z_k)\| = 0.$$

For $e_k = [2+\log(k+1)]^{-1}$, let $R = \|z_0 - z^*\|$ and define

$$K_r = \left\lceil \exp\left(\max\left\{6, \frac{6}{1-r} - 2\right\}\right)\right\rceil, \qquad A = L^2R^2 4^{K_r}\left[\frac{1}{6} + \frac{72\exp(100)}{1-r}\right].$$

Then, for every $N \geq 0$,

$$\min_{0\leq k\leq N} \|F(z_k)\|^2 \leq A\,\frac{2+\log(N+1)}{N+1}.$$

The iteration does not use r, R, K_r or A.

The constant is uniform for fixed L, r and an initial solution-distance bound, but becomes extremely large as $r \uparrow 1$. It is not a practical or optimal complexity estimate. The next sections give a complete proof, including the arbitrary finite prefix and the zero-distance case.

For $k \geq 1$, the final energy and scalar normalization used throughout are

$$\begin{aligned} C(e) &= \frac{(1-e)^2(2-e)}{1+e}, \\ E_k &= \|\bar{z}_k - z^*\|^2 + C(e_k)\|z_k - z_{k-1}\|^2, \\ v_k &= e_{k-1} - e_{k+1}, \qquad \Pi_1 = 1, \qquad \Pi_{k+1} = \Pi_k(1+400v_k), \qquad V_k = E_k/\Pi_k. \end{aligned} \tag{2}$$

All references to z^* in the energy are for analysis only.

2 Normalization and scalar estimates

Write $x^* = z^*$, $g_k = G(z_k)$, and $b_k = \bar{z}_k$. Since $G = F/L$, the normalized assumptions are

$$\begin{aligned} \|G(x) - G(y)\| &\leq \|x - y\|, \qquad G(x^*) = 0, \\ \langle G(x), x - x^*\rangle &\geq -\frac{r}{2}\|G(x)\|^2, \qquad r = \rho L \in [0,1). \end{aligned}$$

In particular, $\|G(x)\| \leq \|x - x^*\|$. The normalized weak-Minty coefficient is $r/2$.

For $0 \leq e \leq 1/2$, put

$$a(e) = (1-e)^2, \quad \beta(e) = \frac{e}{1+e}, \quad C(e) = \frac{(1-e)^2(2-e)}{1+e}.$$

The continuous extension at zero has $C(0) = 2$, while $C(1/2) = 1/4$. Direct differentiation gives

$$C'(e) = \frac{-7 + 8e + e^2 - 2e^3}{(1+e)^2}.$$

On $[0, 1/2]$ this derivative is nonpositive and has absolute value at most 7. Indeed, its numerator is negative because $8e + e^2 \le 17/4 < 7$, and the magnitude of its negative numerator is at most 7. Consequently

$$\frac{1}{4} \le C(e) \le 2, \qquad 0 \le C(q) - C(t) \le 7(t-q) \quad (0 \le q \le t \le 1/2). \tag{3}$$

When $e \le 1/8$, a useful stronger bound is

$$C(e) \ge C(1/8) = \frac{245}{192} > 1. \tag{4}$$

3 Finite prefixes and scalar products

Every update involves evaluation of a globally defined map, arithmetic, and division by $1 + e_k > 0$; thus every finite iterate exists. The canonical initialization gives $b_0 = z_0$ exactly. Let $R = \|z_0 - x^*\|$. Inductively, b_k is a convex combination of z_k and b_{k-1}, and $a(e_k) \le 1$, so

$$\|b_k - x^*\| \le 2^k R, \qquad \|z_{k+1} - x^*\| \le \|b_k - x^*\| + a(e_k)\|g_k\| \le 2^{k+1} R. \tag{5}$$

The induction starts from $b_{-1} = z_0$ and $b_0 = z_0$. This proves both prefix bounds for every $k \ge 0$, including $k = 0$. It also covers $R = 0$: then the complete orbit is the fixed solution and all quantities and bounds are zero.

For $k \ge 1$, $v_k = e_{k-1} - e_{k+1} \ge 0$. For $n \ge 2$,

$$\sum_{k=1}^{n-1} v_k = e_0 + e_1 - e_{n-1} - e_n \le 1.$$

Using $1 + x \le \exp(x)$ gives

$$1 \le \Pi_n = \prod_{k=1}^{n-1} (1 + 400 v_k) \le \exp(400), \tag{6}$$

and this also holds at $n = 1$ because $\Pi_1 = 1$. No product begins at the undefined index zero.

4 Pointwise control before boundedness

Fix any $k \geq 1$ with $p := e_{k-1} \leq 1/8$. Let $t = e_k \leq p$, $h = g_{k-1}$, $b = b_k$, and $d = z_k - z_{k-1}$. The actual recurrence, with its actual retained memory, implies

$$b - z_k = \frac{a(p)}{1+t} h =: Hh, \qquad z_{k-1} - x^* = b - x^* - d - Hh. \tag{7}$$

Apply weak Minty at z_{k-1} and substitute (7):

$$\left(H - \frac{r}{2}\right) \|h\|^2 \leq \langle h, b - x^* - d \rangle. \tag{8}$$

Here

$$H \geq \frac{(1-p)^2}{1+p} \geq \frac{49}{72}, \qquad H - \frac{r}{2} \geq \frac{13}{72}.$$

By (4), $\|b - x^*\|^2 + \|d\|^2 \leq E_k$. Therefore, Cauchy–Schwarz in the product space gives

$$\|b - x^*\| + \|d\| \leq \sqrt{2E_k}, \qquad \|h\| \leq \frac{72\sqrt{2}}{13} \sqrt{E_k} \leq 8\sqrt{E_k}. \tag{9}$$

For $h = 0$ the last inequality is immediate; otherwise it follows by division in (8). No division by E_k or assumption about the orbit is involved. Finally $0 < H \leq 1$ and (7) show

$$\|z_k - x^*\| \leq \|b - x^*\| + H\|h\| \leq 9\sqrt{E_k}, \qquad \|g_k\| \leq 9\sqrt{E_k}. \tag{10}$$

This proves all pointwise estimates, including zero energy cases, before boundedness is established.

5 An identity for the energy calculation

Take arbitrary $t \in (0, 1/2]$, and write

$$\phi = 1 + t, \quad a = (1-t)^2, \quad \beta = t/\phi, \quad \Delta = g - h, \quad s_0 = (a/\phi)h - ag = -(a/\phi)(\Delta + tg).$$

Define

$$D = \beta a \left(\frac{a}{\phi} - r\right), \qquad Q = \beta a r \|g\|^2 - \frac{2\beta a^2}{\phi} \langle g, h \rangle + \beta^2 a^2 \|g\|^2.$$

Then the following is an exact identity:

$$Q + D\|g\|^2 = -\frac{2a}{\phi t} \langle \Delta, s_0 \rangle - \frac{1-t}{t} \|s_0\|^2 - \frac{a^2}{\phi t} \|\Delta\|^2. \tag{11}$$

To verify it analytically, the left side simplifies to

$$\frac{a^2}{\phi^2}\left[t(t-1)\|g\|^2 + 2t\langle g, \Delta\rangle\right].$$

Substitution of $s_0 = -(a/\phi)(\Delta + tg)$ on the right gives the same expression: the coefficient of $\|\Delta\|^2$ is proportional to $2 - (1-t) - \phi = 0$, the mixed coefficient is $2t$, and the coefficient of $\|g\|^2$ is $t(t-1)$.

Young's inequality in (11) gives

$$Q + D\|g\|^2 \le -B\|s_0\|^2 + C(t)\|\Delta\|^2, \tag{12}$$

where

$$B = \frac{1-t}{t} - \frac{a}{\phi t} = \frac{2(1-t)}{1+t}, \quad \frac{a-a^2}{\phi t} = C(t), \quad B - C(t) = \frac{t(1-t)(3-t)}{1+t} \ge 0. \tag{13}$$

No small-t limit or approximation is used.

6 The varying parameter inequality

Fix $k \ge 1$ satisfying

$$e_{k-1} \le \min\{1/8, (1-r)/6\}. \tag{14}$$

Set

$$p = e_{k-1}, \quad t = e_k, \quad q = e_{k+1}, \quad w = p - t, \quad u = t - q, \quad v_k = w + u.$$

Thus $0 \le u, w \le 1/8$. Use the actual state variables

$$b = b_k, \quad g = g_k, \quad h = g_{k-1}, \quad d = z_k - z_{k-1}, \quad E = E_k.$$

Let $a = (1-t)^2$, $\phi = 1 + t$, $\beta = t/\phi$ as above. The deviation in the previous step size is

$$\delta = \frac{a - a(p)}{\phi} = \frac{(p-t)(2-p-t)}{1+t}, \qquad 0 \le \delta \le 2w. \tag{15}$$

The actual recurrence yields exactly

$$b - z_k = (a/\phi - \delta)h, \quad s := z_{k+1} - z_k = s_0 - \delta h, \quad b_{k+1} = b - \beta(q)ag. \tag{16}$$

Introduce only the auxiliary vector $B_0 = b - \beta ag$ for this calculation; it is not a new iterate and no algorithmic restart is made. Weak Minty at z_k, expansion of $\|B_0 - x^*\|^2$, and (16) imply

$$\|B_0 - x^*\|^2 \le \|b - x^*\|^2 + Q + 2\beta a\delta\langle g, h\rangle.$$

In (12), $\|\Delta\| = \|g - h\| \le \|d\|$ by the 1-Lipschitz property. Together with $B \ge C(t)$ from (13), this gives

$$\begin{aligned}\|B_0 - x^*\|^2 + C(t)\|s\|^2 \le E - D\|g\|^2 + 2\beta a\delta\langle g, h\rangle \\ - 2C(t)\delta\langle s_0, h\rangle + C(t)\delta^2\|h\|^2.\end{aligned} \tag{17}$$

The pointwise estimates already proved, and (4), give

$$\|b - x^*\| \le \sqrt{E}, \quad \|d\| \le \sqrt{E}, \quad \|h\| \le 8\sqrt{E}, \quad \|g\| \le 9\sqrt{E}. \tag{18}$$

Using $s_0 = -(a/\phi)(\Delta + tg)$, $a/\phi \le 1$, $t \le 1/8$, and (15), we further have

$$\|s_0\| \le (1 + 9/8)\sqrt{E} \le 3\sqrt{E}, \quad \|B_0 - x^*\| \le 3\sqrt{E}, \quad \|s\| \le (3 + 16w)\sqrt{E} \le 5\sqrt{E}. \tag{19}$$

All bounds in (18)–(19) are inequalities that remain valid when $E = 0$.

The three error terms in (17) are bounded, individually, by

$$\begin{aligned}2\beta a\delta|\langle g, h\rangle| &\le 2(1/8)(2w)(9)(8)E = 36wE, \\ 2C(t)\delta|\langle s_0, h\rangle| &\le 2(2)(2w)(3)(8)E = 192wE, \\ C(t)\delta^2\|h\|^2 &\le 2(2w)^2(8)^2E = 512w^2E \le 64wE.\end{aligned}$$

Consequently

$$\|B_0 - x^*\|^2 + C(t)\|s\|^2 \le E - D\|g\|^2 + 292wE. \tag{20}$$

The averaging change from t to q is described by

$$\varepsilon = a\big(\beta(t) - \beta(q)\big), \qquad 0 \le \varepsilon \le u, \quad b_{k+1} = B_0 + \varepsilon g, \tag{21}$$

because $\beta'(e) = 1/(1+e)^2 \le 1$ and $a \le 1$. Equations (18)–(19) bound its contribution by

$$\|b_{k+1} - x^*\|^2 - \|B_0 - x^*\|^2 \le 54uE + 81u^2E \le 65uE. \tag{22}$$

Finally, changing the energy's scalar weight from $C(t)$ to $C(q)$ contributes at most

$$[C(q) - C(t)]\|s\|^2 \le 7u \cdot 25E = 175uE. \tag{23}$$

Adding (20), (22), and (23) proves the stronger intermediate estimate

$$E_{k+1} \le [1 + 292w + 240u]E_k - D\|g_k\|^2. \tag{24}$$

To bound D, let $\gamma = 1 - r > 0$. The threshold gives $t \le \gamma/6$, and

$$\frac{a}{\phi} = \frac{(1-t)^2}{1+t} \ge 1 - 3t, \quad \frac{a}{\phi} - r \ge \gamma - 3t \ge \frac{\gamma}{2}, \quad \frac{a}{\phi} \ge \frac{49}{72} \ge \frac{1}{2}.$$

It follows that

$$D = t\frac{a}{\phi}\left(\frac{a}{\phi} - r\right) \geq \frac{\gamma}{4}t \geq \frac{\gamma}{12}t. \tag{25}$$

Since $292w + 240u \leq 400(w+u) = 400v_k$, (24)–(25) prove exactly the announced inequality

$$E_{k+1} \leq (1 + 400v_k)E_k - \frac{1-r}{12}e_k\|G(z_k)\|^2. \tag{26}$$

Dividing by $\Pi_{k+1} = \Pi_k(1 + 400v_k)$ gives explicitly

$$V_{k+1} \leq V_k - \frac{(1-r)e_k}{12\Pi_{k+1}}\|G(z_k)\|^2. \tag{27}$$

Plateaus cause no difficulty: either or both of w, u may vanish, and the corresponding error terms vanish exactly.

7 Boundedness and weighted residual summability

Because $e_k \to 0$, there is a finite integer $K \geq 1$ such that every $k \geq K$ satisfies the threshold. Define the relative tail product

$$P_K = 1, \qquad P_n = \prod_{k=K}^{n-1}(1 + 400v_k) = \frac{\Pi_n}{\Pi_K} \quad (n > K).$$

For $n > K$, telescoping the scalar variations gives

$$\sum_{k=K}^{n-1} v_k = e_{K-1} + e_K - e_{n-1} - e_n \leq e_{K-1} + e_K \leq \frac{1}{4}.$$

Hence

$$1 \leq P_n \leq \exp(100). \tag{28}$$

This stronger tail bound, rather than the global $\exp(400)$, is what produces the asserted constant in the rate bound. Dividing (26) by P_{k+1} and summing gives, for all $n \geq K$,

$$\frac{E_{n+1}}{P_{n+1}} + \frac{1-r}{12}\sum_{k=K}^{n}\frac{e_k\|g_k\|^2}{P_{k+1}} \leq E_K.$$

All terms are nonnegative. In particular,

$$E_n \leq \exp(100)E_K \quad (n \geq K), \qquad \sum_{k=K}^{\infty} e_k\|g_k\|^2 \leq \frac{12\exp(100)}{1-r}E_K. \tag{29}$$

The energy directly bounds $b_n - x^*$; (10) bounds $z_n - x^*$ for the same tail. The finite prefix is bounded by (5), so both actual sequences are bounded. The finitely many residual terms before K are finite, and $F = LG$, so (29) yields

$$\sum_{k=0}^{\infty} e_k \|F(z_k)\|^2 < \infty.$$

Since $\sum e_k = \infty$, a strictly positive eventual lower bound on $\|F(z_k)\|$ would contradict this finite sum. Thus $\liminf \|F(z_k)\| = 0$. No point-convergence or all-iterate residual-convergence assertion is needed or made.

8 A logarithmic schedule and its constant

For $e_k = 1/[2 + \log(k+1)]$, $e_0 = 1/2$, positivity and monotonicity are immediate, and $e_k \to 0$. For every $N \geq 0$,

$$\sum_{k=0}^{N} e_k \geq \frac{N+1}{2 + \log(N+1)} \longrightarrow \infty, \tag{30}$$

so the divergent-sum condition holds.

Set

$$K = K_r = \left\lceil \exp\left(\max\left\{ 6, \frac{6}{1-r} - 2 \right\} \right) \right\rceil.$$

Then $K \geq 1$, and for every integer $k \geq K$,

$$e_{k-1} = \frac{1}{2 + \log k} \leq \frac{1}{8}, \qquad e_{k-1} \leq \frac{1-r}{6}. \tag{31}$$

The shift is correct: condition (14) checks e_{k-1}, hence the logarithm in (31) is $\log k$, and the ceiling in the displayed K_r suffices.

The coarse finite-prefix bounds (5) and $C \leq 2$ give

$$\begin{aligned} E_K &\leq 4^K R^2 + 2(2^K + 2^{K-1})^2 R^2 \\ &= \frac{11}{2} 4^K R^2 \leq 6\, 4^K R^2. \end{aligned} \tag{32}$$

For the residual prefix, using $e_k \leq 1/2$ and $\|g_k\| \leq 2^k R$,

$$\sum_{k=0}^{K-1} e_k \|g_k\|^2 \leq \frac{R^2}{2} \sum_{k=0}^{K-1} 4^k = \frac{R^2(4^K - 1)}{6} \leq \frac{4^K R^2}{6}. \tag{33}$$

Combine (29), (32), and (33), multiplying by L^2, to obtain

$$\sum_{k=0}^{\infty} e_k \|F(z_k)\|^2 \leq L^2 R^2 4^{K_r} \left[\frac{1}{6} + \frac{72 \exp(100)}{1-r} \right] = A. \tag{34}$$

For every $N \geq 0$, the minimum is at most the weighted average, and (30)–(34) give

$$\min_{0\leq k\leq N} \|F(z_k)\|^2 \leq \frac{\sum_{k=0}^{N} e_k \|F(z_k)\|^2}{\sum_{k=0}^{N} e_k} \leq A\frac{2+\log(N+1)}{N+1}.$$

This includes $N = 0$, all $N < K_r$, and $R = 0$; the proof never restricts the minimum to the tail or divides by R.

9 Scope and ancestry

The fixed-parameter GRAAL geometry and its relation to optimistic methods are established in [1]. Generalized optimistic and adaptive extragradient methods under weak Minty are studied in [2]; methods based on inexact proximal calculations cover the full admissible weak-Minty range with explicit oracle accounting [3]. Those results concern different parameter choices or implemented recurrences. The result here is a particular automatic schedule for the original two GRAAL lines, retaining their averaging memory and using a known global Lipschitz bound. No first-parameter-free-method claim is made.

The argument uses the displayed fixed-parameter admissibility condition in [1, Theorem 7] as motivation, and proves its own coupled-step estimates directly. The energy coefficient motion is controlled by the actual schedule differences, including plateaus. The late threshold identifies a finite tail only in the proof; the method neither knows nor detects its onset.

The result gives boundedness and a bound on the smallest residual square. It makes no assertion of convergence of every residual or of the points. The assumptions do not include unknown L, constraints, proximal terms, stochasticity or noise. The schedule is predetermined and does not estimate the weak-Minty parameter from data.

References

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